CatSight

Event-aware contract intelligence for catastrophe reinsurance

When a typhoon or earthquake hits, CatSight tells a reinsurer **which treaties respond, what each layer pays, and which contract clause could change that answer** — with every quote machine-verified against the page.

Live: **https://catsight-api-342162443546.us-central1.run.app**
Code: **https://github.com/venketraj/catsight-ai-buildercup-2026**

Indicative first view on synthetic data. Not a catastrophe-model output. Not underwriting or claims advice.

THE PROBLEM

The morning after a typhoon

A catastrophe analyst at a reinsurer needs a **defensible first view of their own exposure** — by treaty, by layer — before industry figures exist.

The first number matters, it is hard, and it moves. **Typhoon Jebi, 2018:**

- 857,284 claims, ¥1,067.8bn paid — the largest ever for a Japanese typhoon GIAJ
- First modelled range $3–5.5bn, about 10 days after landfall Moody's RMS
- Insured loss later about $13bn — roughly 3× the early figure Swiss Re

Industry figure	Arrives
Modeller event response	**4–20 days**
PERILS (Japan typhoon)	**6 weeks**
Verisk PCS Japan	**up to 90 days**
CatSight, on the reinsurer's own book	**seconds**

Sources: providers' published event-response releases. Industry figures describe the market, not one reinsurer's treaties.

WHY IT IS HARD TODAY

The work is manual, and the wording is where it breaks

Step	Today	Pain
Event intelligence	Licensed modeller feeds	Expensive; describes the market, not your contracts
Exposure overlay	Cat-modelling and exposure tools	Specialist time, batch runs under pressure
Contract application	Spreadsheets	Error-prone when rushed
Wording check	Underwriting, claims and legal re-read PDFs	**Slow; endorsements get missed; disputes surface months later**
Reporting	Flash estimates to the board	Must be fast and defensible

48% of insurers do not license catastrophe models, and **nearly 60%** run cat teams of five or fewer Aon 2025. Treaty structures "might only exist in their capital model, a homegrown tool, or even in a spreadsheet" Moody's.

WHAT EXISTS VS WHAT IS NEW

CatSight is the join

Category	Examples	Does well	Stops at
Cat-model event response	Moody's RMS, Verisk	Hazard footprints, modelled loss, coded treaty terms	Your treaty **wording**
AI contract review	Document-AI platforms	Clause extraction, citations	The **event** and the layer maths
Generic LLM over PDFs	Chat with a contract	Fast answers	**Verifiable numbers**, audit trail
CatSight		**event × exposure × layers × wording — with verified citations and tested maths**	

Complementary, not a replacement. CatSight is not a cat model: a licensed footprint can replace the public hazard data behind the same interface. To our knowledge, event-response products stop at the modelled loss and AI wording tools stop at the document.

ARCHITECTURE

One Cloud Run service, two ADK graphs

Browser
React UI, served by the same service · SSE stream

▼

Cloud Run — FastAPI
ADK 2.11 analysis graph · Courtroom graph · deterministic loss engine · guards

▼

Vertex AI
Gemini 3.5 Flash-Lite · 3.8 Flash · embedding-001

Firestore
treaty book · 252 clause vectors · runs

Cloud Storage
treaty PDFs

Cloud Logging · Trace · Monitoring · Cloud Build · dedicated service account, no API keys

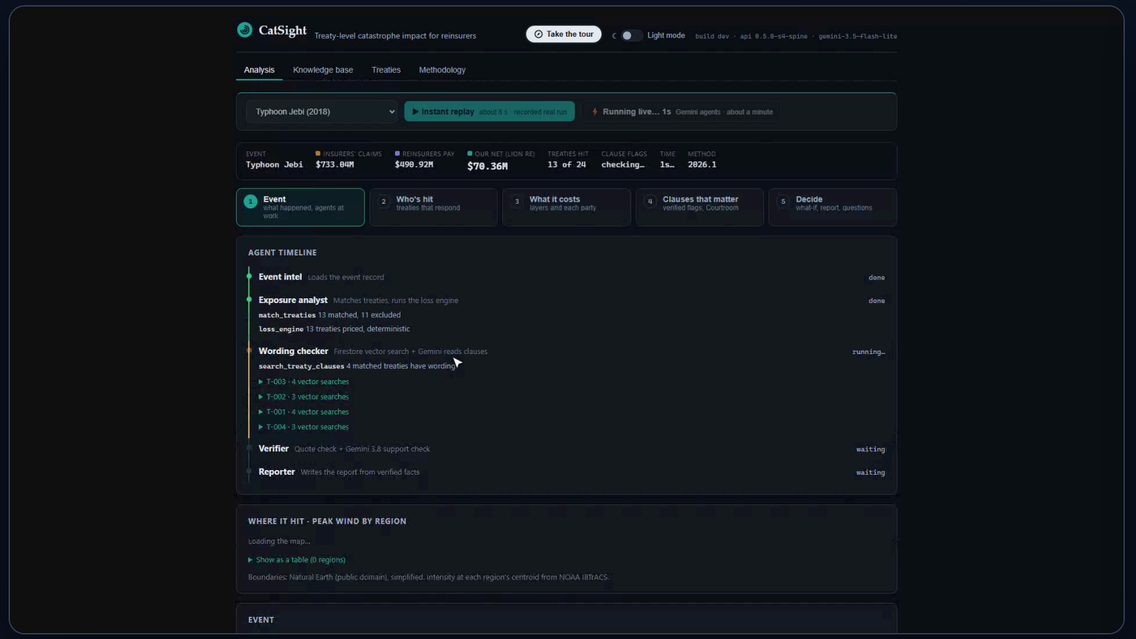
Analysis
event_intel → exposure_analyst (no model) → wording_checker (vector search) → verifier (loop) → reporter

Clause Courtroom
brief → cedent_counsel ‖ reinsurer_counsel (parallel, blind) → quote check → price (engine) → arbiter (does not decide)

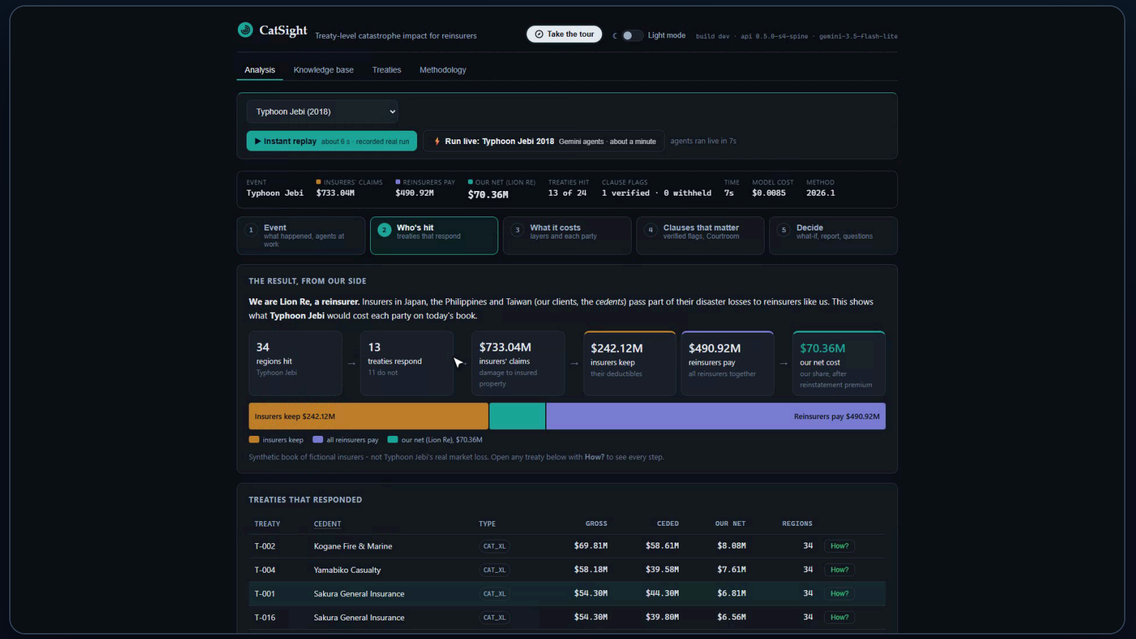
Gemini reads. Tested code calculates. Every figure is deterministic Python; every figure in AI-written text is checked against it.

THE PRODUCT — LIVE

Agents stream, the money lands first



Agents working live: retrieval per treaty, then verification



Who pays what: 13 of 24 treaties respond; insurers, reinsurers and our net

THE MONEY MOMENT — ONE CLAUSE, ONE NUMBER

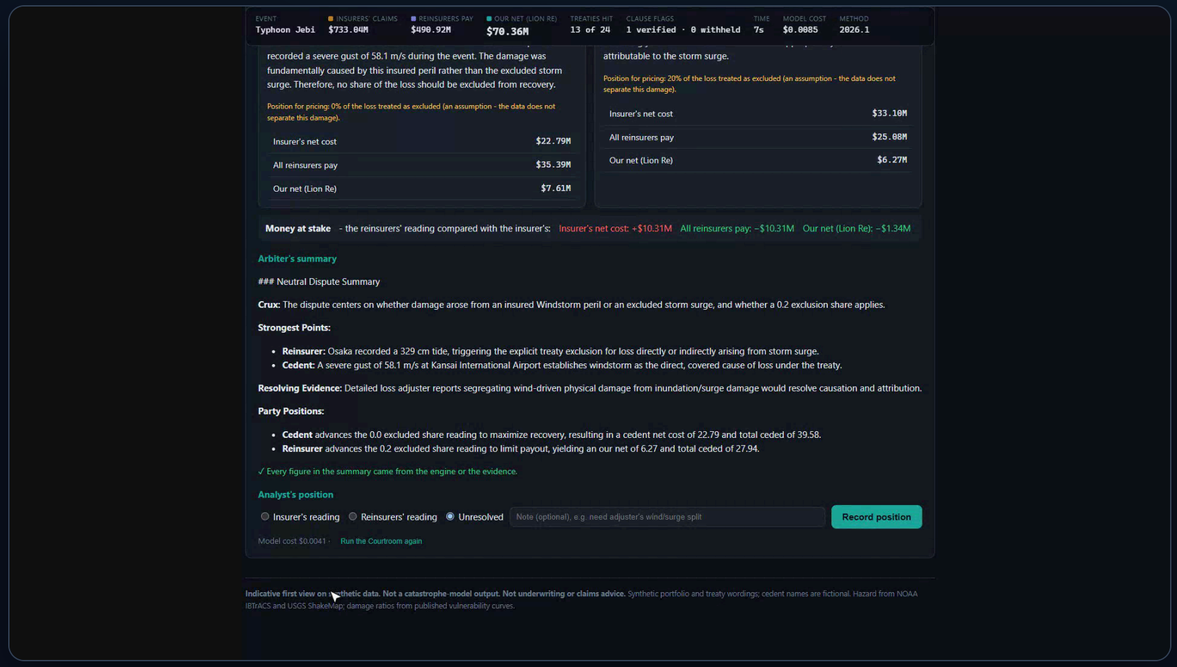
Jebi, treaty T-004: the storm-surge exclusion

Jebi drove a **record 329 cm tide** into Osaka Bay. T-004 (p. 2, cl. 8, verified) excludes storm surge.

Two AI counsel argue in parallel. Each quote is checked; each reading is **priced by the engine**. The surge share is a stated assumption — no data separates wind from water.

Recorded live run	Move
Total ceded (20% surge share)	**−$11.64M**
Insurer's net cost	**+$10.31M**
All reinsurers pay	**−$10.31M**
Our net (Lion Re)	**−$1.34M**

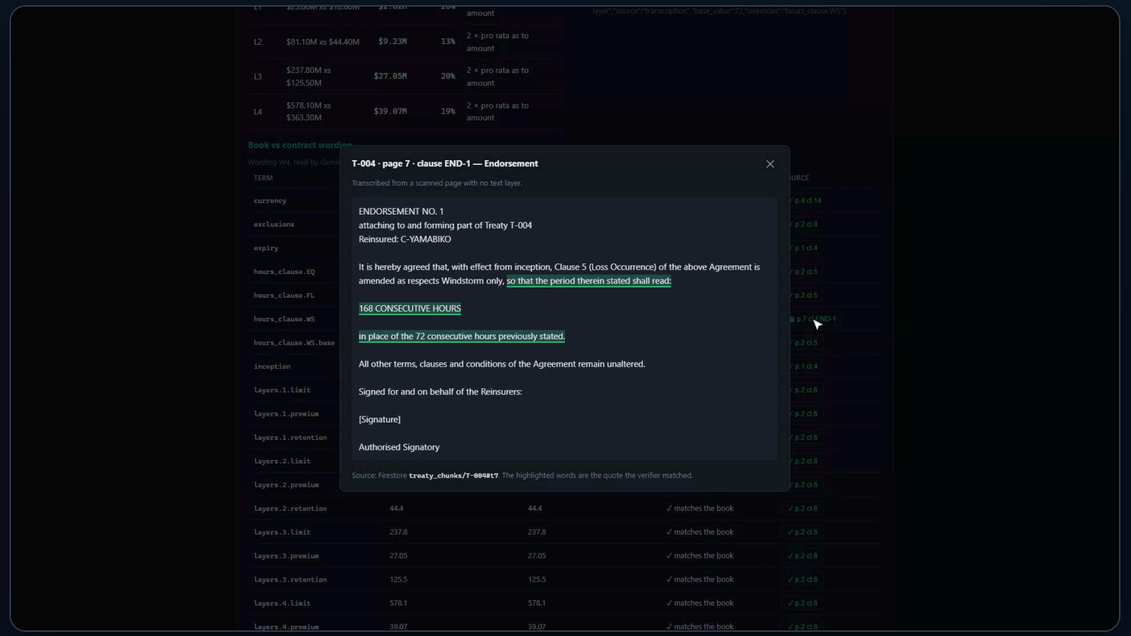
What the reinsurers save, the insurer carries. Counsel choose their share each run, so figures vary by run. Issue-spotting, not legal advice — the analyst decides.



TRUST ARCHITECTURE

Why a finance team can let an LLM near a loss report

- **The model never calculates.** Every figure from tested `Decimal` Python — 870 tests.
- **Every wording claim is cited** to treaty, page and clause.
- **Every citation is machine-checked**: exact quote match, then a model support check; one revision, then withheld.
- **Withheld, visibly**: "n of n verified · k withheld", with reasons.
- **Scanned pages** are transcribed by Gemini and badged differently.
- **A human decides.** The arbiter summarises; the analyst records a position.



The endorsement detective: Gemini reads a scanned page with no text layer — 168 hours replaces the body's 72

RESULTS — MEASURED

What we measured, and what we did not

0.7 s
money on screen (p50, 5 live runs)

9.8 s
complete with verified flags (p50; 8–18 s)

$0.010
model cost per analysis (p50)

0
report figures not traceable to the engine

Measure	Result	Honest caveat
Loss engine vs hand-checked worked example	**exact**	unit-tested on every figure
Treaty-term extraction	**99.5%** (185/186)	upper bound: prompts developed on these 8 wordings; no blind set yet
Citations	**186/188** cited · **185/186** check out	per-field audit published
Storm-surge flag on Jebi	raised 8/8 · verified 7/8	severity varies (HIGH 4/8); the Courtroom now argues HIGH or MEDIUM
Not measured	timed manual baseline · usability · flag recall over a labelled corpus · blind extraction set	

BUILT ON GOOGLE CLOUD

Every service, and why

Service	Role
Gemini 3.5 Flash-Lite on Vertex AI (global)	Wording checks, counsel, reports — cheap generation
Gemini 3.8 Flash	Verification and the arbiter — stronger judging
Google ADK 2.11 graph `Workflow`	A verification loop and a parallel Courtroom fan-out — not a linear chain
Firestore + vector search, `gemini-embedding-001` 768-d	Treaty book, 252 clause vectors with a treaty pre-filter, completed runs
Cloud Run	One service: API, SSE stream and UI; max 3 instances, per-visitor and daily caps
Cloud Logging · Trace · Monitoring	Structured guardrail events; per-agent ADK spans; a dashboard of latency, cost, citations and Courtroom outcomes
Cloud Storage · Cloud Build	Treaty PDFs; source deploys

IMPACT AND SCALE

Faster, less-disputed recoveries

Reinsurance recoveries are what let insurers pay households and small businesses after a disaster.

- **Effort (estimate):** 30–60 analyst-hours per event → 3.5–8 hours of review; about 165–675 hours a year for a mid-size reinsurer. A timed baseline is next.
- **Wording risk:** on the worked example, one hours clause moves a layer by up to 80%.
- **Market:** Singapore writes S$27.6bn of reinsurance premium and hosts 16 of the top 25 reinsurers' regional hubs; 91% of APAC's 2023 economic catastrophe losses were uninsured MAS.

Decision support, not a payout engine.

Path to production

- Licensed cat-model footprints behind the hazard interface
- Live event intake from USGS / GDACS alerts
- Authenticated upload of real treaty packs; BigQuery exposure
- Multi-tenancy; A2A / MCP so a broker's agent can call CatSight
- Loss development as advices arrive

TEAM, ASK, NEXT STEPS

Techno Crackers

Venket Raj S · Malik Chandra Pandian · Arun Veerappan · Nagaraj

Reinsurance domain knowledge and engineering, building with Gemini and Google ADK.

Next: pilot on an anonymised treaty pack · a timed manual baseline · a blind extraction set · pre-landfall mode on forecast tracks.

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Jebi, Hagibis, Haiyan and Noto were real events in which people died. Synthetic insurers and treaties; public hazard data: NOAA IBTrACS, USGS ShakeMap, Natural Earth.